

Lec 32

Concentration Inequalities.

Recap.

① Markov inequality

X is a non-negative R.V.

$$P(X \geq a) \leq \frac{E[X]}{a} \quad \text{for any } a > 0.$$

Example:

② $X \sim \text{Binomial}(n, p)$

$$P(X > \alpha n) \leq \frac{np}{n\alpha} = \frac{p}{\alpha}.$$

$$0 < p < \alpha < 1, \quad \text{For } p = 1/2, \quad \alpha = 3/4.$$

$$P(X > \alpha n) \leq 2/3.$$

② Chebyshev inequality.

X is a R.V with finite mean μ & variance, σ^2

$$P(|X - \mu| > \epsilon) \leq \frac{\text{Var}(X)}{\epsilon^2}.$$

$$P(X \notin [\mu - \epsilon, \mu + \epsilon])$$

$$Y = (X - \mu)^2.$$

LHS

$$= P(Y > \epsilon^2) \leq \frac{E[Y]}{\epsilon^2} = \frac{\text{Var}(X)}{\epsilon^2}.$$

Markov
inequality

$$\textcircled{a} \quad X \sim \text{Binomial}(n, p)$$

$$P(X > \alpha n) = P(X - pn > (\alpha - p)n) \\ \leq P(|X - pn| > (\alpha - p)n)$$

$$\leq \frac{\text{Var}(X)}{(\alpha - p)^2 n^2} = \frac{\cancel{p(1-p)}}{(\alpha - p)^2 \cancel{n^2}}$$

$$= \frac{p(1-p)}{(\alpha - p)^2} \frac{1}{n}$$

For $p = 1/2$, $\alpha = 3/4$

$$P(X > \alpha n) \leq \frac{4}{n}$$

⑥ WLLN (Weak law of large numbers)

$$S_n = \underbrace{X_1 + \dots + X_n}_n \quad X_i \text{'s are iid}$$

How well does S_n approximate $E[X_1]$.

$$E[S_n] = E[X_1]$$

$$\text{Var}[S_n] = \frac{\text{Var}(X_1)}{n}$$

$$P(|S_n - E[X_1]| > \varepsilon) \leq \frac{\text{Var}(S_n)}{\varepsilon^2} = \frac{\text{Var}(X_1)}{n\varepsilon^2}$$

$$S_n \notin [E[X_1] - \varepsilon, E[X_1] + \varepsilon]$$

$$\lim_{n \rightarrow \infty} P(|S_n - E[X_1]| > \varepsilon) = 0$$

③ Chernoff bound.

$$P(X > a) = P(e^{Xt} > e^{at}).$$

for any $t > 0$.

$$\text{let } Y = e^{Xt}.$$

$$= P(Y > e^{at})$$

as Y is
a non-negative $\leq \frac{E[Y]}{e^{at}}$

$$P(X > a) \leq \frac{E[e^{Xt}]}{e^{at}} \quad \text{for any } t > 0.$$

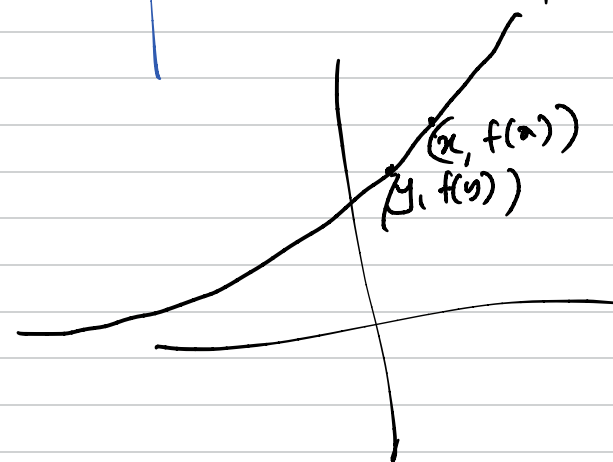
$$P(X > a) \leq \inf_{t > 0} \frac{E[e^{Xt}]}{e^{at}}$$

$$P(X > a)$$

$$= P(f(X) > f(a))$$

if f is a monotonically
increasing function

if $x > y$ then
 $f(x) > f(y)$



$$= \inf_{t>0} M_X(t) e^{-at}.$$

$$= \inf_{t>0} e^{\ln M_X(t) - at}.$$

$$= \inf_{t>0} e^{-[at - \ln M_X(t)]}.$$

$$= e^{-\sup_{t>0} [at - \ln M_X(t)]}.$$

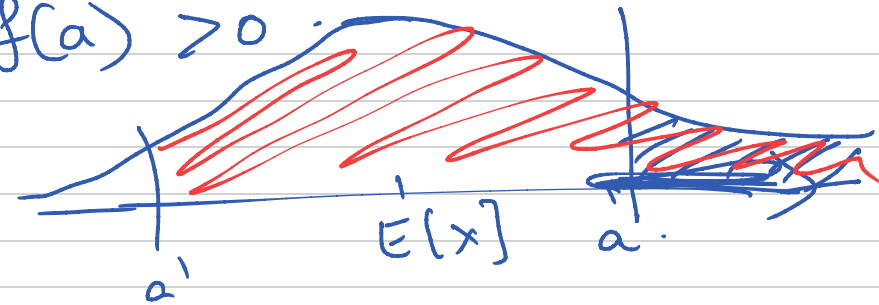
$$f_a(t) = at - \ln M_X(t).$$

$$f(a) = \sup_{t>0} f_a(t).$$

$$\boxed{P(X > a) \leq e^{-f(a)}}.$$

"This bound is meaningful if $e^{-f(a)} \leq 1$
 $\Leftrightarrow f(a) \geq 0$.

Lemma: If $a > E[x]$ then $f(a) > 0$.



$$f_a(t) = at - \ln M_x(t).$$

$$f_a = \sup_{t>0} f_a(t)$$

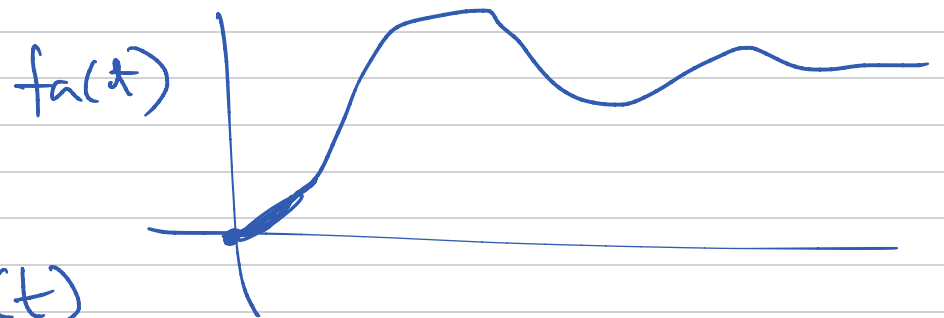
$$M_x(t) = E[e^{xt}]$$

$$M_x(0) = E[e^{x \cdot 0}]$$

$$= 1$$

$$f_a(0) = a \cdot 0 - \ln M_x(0) = a \cdot 0 - \ln 1 = 0.$$

"We want to show that $f_a(t)$ is an increasing function around $t=0$ ".



$$\frac{\partial f_a(t)}{\partial t} = a - \frac{1}{M_x(t)} \frac{\partial M_x(t)}{\partial t}.$$

$$\left. \frac{\partial f_a(t)}{\partial t} \right|_{t=0} = a - \frac{1}{M_x(0)} \left. \frac{\partial M_x(t)}{\partial t} \right|_{t=0} = a - E[x] > 0.$$

$\Rightarrow f_a(t)$ is an increasing function around $t=0$.

$$\therefore f(a) = \sup_{t \geq 0} f_a(t) > 0.$$

Examples :

① $X \sim N(0, 1)$.

$$P(X \geq a) \leq e^{-f(a)}.$$

$$f(a) = \sup_{t \geq 0} f_a(t)$$

$$f_a(t) = at - \ln M_X(t).$$

for $X \sim N(0, 1)$, $M_X(t) = e^{t^2/2}$.

$$\Rightarrow f_a(t) = at - t^2/2.$$

$$f(a) = \sup_{t \geq 0} at - t^2/2.$$

$$f'_a(t^*) = a - t^* = 0.$$

$$\underline{t^* = a}$$

$$f''_a(t) = -1 < 0.$$

$$f(a) = f_a(t^*) = f_a(a).$$

$$= a \times a - a^2/2 = a^2/2.$$

$$1 - \Phi(a) \leq e^{-a^2/2}$$

$$\underline{\Phi(a) \geq 1 - e^{-a^2/2}}$$

$$\boxed{P(X \geq a) \leq e^{-a^2/2}} \rightarrow \text{Chernoff}$$

$$\underline{P(X \geq a) = P\left(\frac{X-0}{1} \geq a\right)}$$

$$\boxed{P(X \geq a) \leq \frac{\text{Var}(X)}{a^2} = \frac{1}{a^2}} \rightarrow \text{Chebyshev.}$$

② $X \sim \text{Binomial}(n, p)$, $\alpha p < \alpha \leq 1$

$$P(X > \alpha n) \leq e^{-f(a)}$$

where $a = \alpha n$.

$$f(a) = \sup_{t > 0} f_a(t)$$

$$= \sup_{t > 0} at - \ln M_X(t).$$

$$M_X(t) = ((1-p) + pe^t)^n.$$

$$\frac{\partial f_a(t)}{\partial t} = a - \frac{n((1-p) + pe^t)^{n-1} \cdot pe^t}{((1-p) + pe^t)^n}$$

$$= 0.$$

$$\Rightarrow a = \frac{n pe^{t^*}}{(1-p) + pe^{t^*}} \Rightarrow \alpha p = \frac{pe^{t^*}}{(1-p) + pe^{t^*}}.$$

$X_i \sim \text{Bernoulli}(p)$

$$M_{X_i}(t) = (1-p) + pe^t$$

X_i 's are iid
Bernoulli(p)

$$X = X_1 + \dots + X_n.$$

$$M_X(t) = E[e^{\sum X_i}]$$

$$= (M_{X_i}(t))^n$$

$$= ((1-p) + pe^t)^n$$

$$\alpha(1-p) + \alpha p e^{t^*} = p e^{t^*}.$$

$$(1-\alpha) p e^{t^*} = \alpha(1-p).$$

$$e^{t^*} = \frac{\alpha(1-p)}{p(1-\alpha)}.$$

$$t^* = \ln \frac{\alpha(1-p)}{p(1-\alpha)}.$$

check that
 $f''_a(t^*) < 0$.

$$f(a) = a t^* = \ln M_X(t^*).$$

$$= (\alpha n) \ln \frac{\alpha(1-p)}{p(1-\alpha)} - \ln \left[(1-p) + p e^{t^*} \right]^n.$$

$$= (\alpha n) \ln \frac{\alpha(1-p)}{p(1-\alpha)} - \ln \left[(1-p) \left[1 + \frac{\alpha}{1-\alpha} \right] \right]^n \frac{\alpha(1-p)}{(1-\alpha)}.$$

$$= (\alpha n) \ln \frac{\alpha(1-p)}{p(1-\alpha)} - \ln \left(\frac{(1-p)}{(1-\alpha)} \right)^n.$$

$$f(a) = \ln \left[\frac{\alpha(1-p)}{p(1-\alpha)} \right]^{\alpha n} \left[\frac{(1-\alpha)}{(1-p)} \right]^n.$$

$$P(X \geq a) \leq e^{-f(a)}.$$

For $p = 1/2$, $\alpha = 3/4$.

$$\begin{aligned} f(a) &= \ln \left[\frac{3/4 \times 1/2}{1/2 \times 1/4} \right]^{\frac{3}{4}n} \left[\frac{1/4}{1/2} \right]^n. \\ &= \ln \frac{3^{3/4 n}}{2^n} = \ln \frac{27^{n/4}}{16^{n/4}}. \end{aligned}$$

$\alpha n = \frac{3}{4}n$

$$f(a) = \ln \left(\frac{27}{16} \right)^{n/4} = \frac{n}{4} \ln \left(\frac{27}{16} \right).$$

$$P(X \geq a) \leq e^{-\frac{n}{4} \ln \left(\frac{27}{16} \right)}.$$

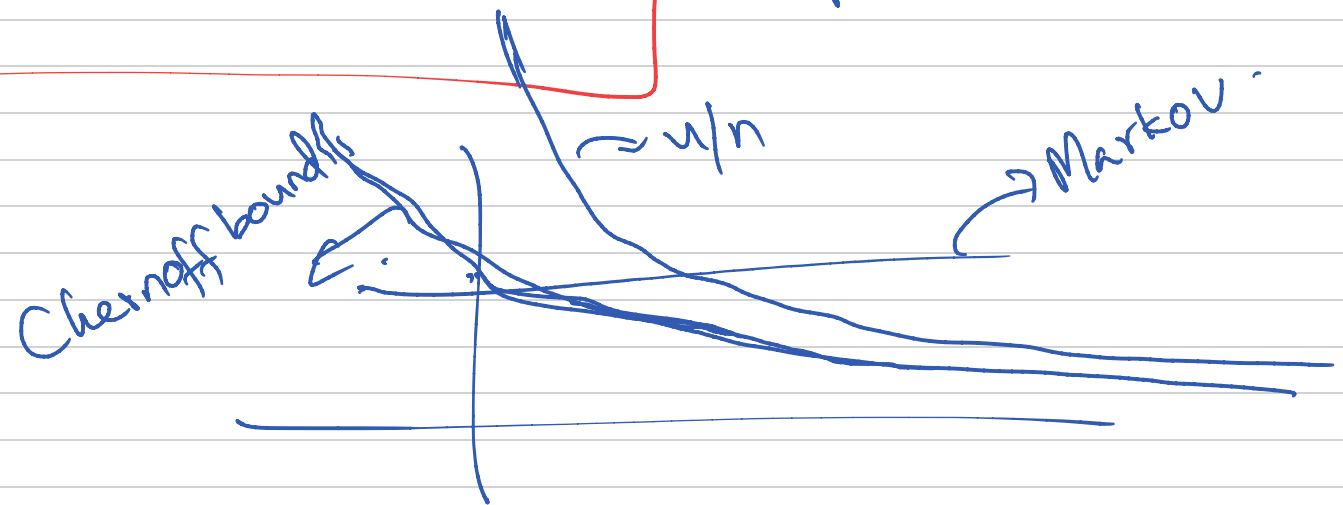
Chernoff bound.

$$P(X \geq a) \leq \frac{4}{n}.$$

Chebyshev

$$P(X \geq a) \leq \frac{3}{n}.$$

Markov.



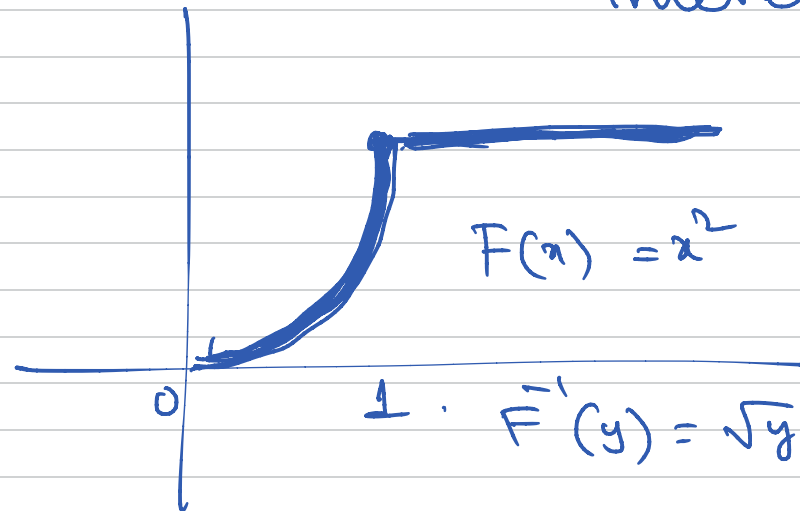
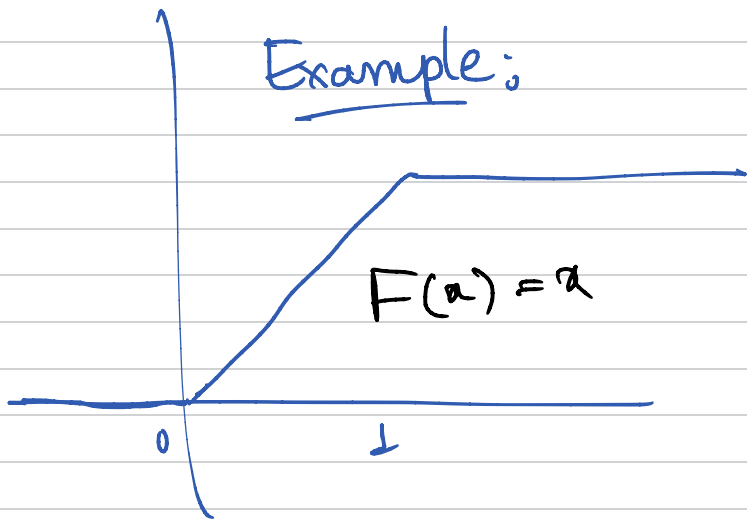
Inverse Sampling

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Chapter 10

① Let $U \sim \text{Uniform}[0, 1]$. Given U , you would like to generate a sample from a distribution that has CDF $F(x)$.

② Say F is invertible in the domain of interest

Example:



Let $Y = F^{-1}(U)$. $U \in [0, 1]$.

$$P(Y \leq y) = P(F^{-1}(U) \leq y).$$

$$= P(U \leq F(y)).$$

$$= F(y)$$

$$\textcircled{a.1} \quad F(x) = 1 - e^{-\lambda x} \quad x > 0.$$

$$Y = F^{-1}(U).$$

$$= \frac{-\ln(1-u)}{\lambda}.$$

$$1 - e^{-\lambda x} = u.$$

$$(1-u) = e^{-\lambda x}$$

$$\boxed{\frac{-\ln(1-u)}{\lambda} = x.}$$

$$\Rightarrow Y \sim \exp(\lambda).$$

Suppose
⑥ X is discrete R.V with PMF $(P_X(x) | x \in \mathbb{Z})$

How to get a sample of X given that you have a sample for uniform R.V ?

① $X \sim \text{Bernoulli}(\frac{1}{2})$.

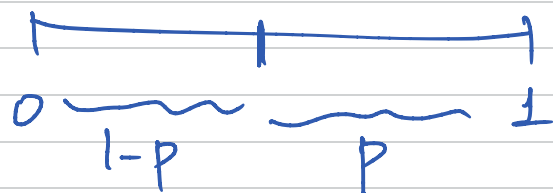
$$X = \begin{cases} 0 & \text{if } u < \frac{1}{2} \\ 1 & \text{if } u \geq \frac{1}{2} \end{cases}$$

$$P(X=0) = P(u < \frac{1}{2}) = \frac{1}{2}.$$

$$P(X=1) = P(u \geq \frac{1}{2}) = \frac{1}{2}$$

(ii) $X \sim \text{Bernoulli}(p)$.

$$X = \begin{cases} 0 & u < (1-p) \\ 1 & u \geq (1-p) \end{cases}$$



$$P(X=0) = P(U < 1-p) = 1-p.$$

$$= P(U \in [0, 1-p))$$

$$P(X=1) = P(U \geq 1-p)$$

$$= P(U \in [1-p, 1]) = p.$$

(iii) X st PMF is given by
 $(p_1, p_2, \dots, p_n), \quad \sum p_i = 1$



$$X = \begin{cases} 1 \\ 2 \\ 3 \\ \vdots \\ i \\ \vdots \end{cases}$$

$$U \in [0, p_1]$$

$$U \in [p_1, p_1+p_2]$$

$$U \in [p_1+p_2, p_1+p_2+p_3]$$

$$U \in \left[\sum_{j=1}^{i-1} p_j, \sum_{j=1}^i p_j \right]$$

(iv) $X \sim \text{Geometric}(p)$. How to sample for
 X using $\text{Uniform}(0,1)$ sample U ?
(Exercise)